

Rocket Nozzle Performance Reconstruction from CAD Geometry

Project Overview

This project began with the design of a Merlin-inspired converging-diverging rocket nozzle in FreeCAD. Rather than performing CFD directly, the objective was to reconstruct and analyze the internal flow field using classical compressible flow theory and numerical methods in Python.

The project bridges CAD geometry and aerospace propulsion analysis by estimating nozzle performance from geometric measurements obtained from the CAD model.

CAD Geometry

The nozzle geometry was designed in FreeCAD and measured directly from the model.

Measured Dimensions

- Chamber Diameter: 536 mm
- Throat Diameter: 260 mm
- Exit Diameter: 460 mm

These dimensions were used to calculate local flow area throughout the nozzle.

Analysis Assumptions

The flow model was based on the following assumptions:

- Steady flow
- One-dimensional flow
- Compressible gas
- Isentropic expansion
- Ideal gas behavior
- Constant specific heat ratio

Operating Conditions

- Stagnation Temperature (T_0): 3000 K
- Chamber Pressure (P_0): 1 MPa (10 bar)
- Specific Heat Ratio (γ): 1.4
- Gas Constant (R): 287 J/kg·K

Governing Equations

Area-Mach Relation

Used to determine Mach number from local area ratio.

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}$$

The equation was solved numerically using SciPy's `fsolve()` function.

Temperature Relation

$$T = \frac{T_0}{1 + \frac{\gamma - 1}{2} M^2}$$

Pressure Relation

$$P = P_0 \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{-\frac{\gamma}{\gamma - 1}}$$

Density Relation

$$\rho = \frac{P}{RT}$$

Velocity Relation

$$V = M \sqrt{\gamma RT}$$

Numerical Method

The nozzle was discretized into multiple axial sections.

For each section:

- Local radius was computed.
- Local area was calculated.
- Area ratio A/A^* was determined.
- Mach number was obtained using numerical root solving.

5. Temperature, pressure, density, and velocity were calculated.

The solver reconstructed the entire flow field along the nozzle length.

Flow Field Results

Expansion Ratio

$$A_e/A_t = 3.106$$

Exit Conditions

- Exit Mach Number: 2.674
- Exit Temperature: 1234.44 K
- Exit Velocity: 1883.35 m/s
- Exit Pressure: 44.7 kPa

Flow Behavior

The nozzle demonstrated physically correct converging-diverging behavior:

- Subsonic flow in chamber
- Sonic flow at throat
- Supersonic expansion in diverging section
- Continuous acceleration to nozzle exit

Generated Flow Profiles

The Python solver successfully reconstructed:

- Mach Number Distribution
- Temperature Distribution
- Velocity Distribution
- Pressure Distribution
- Density Distribution

Observed trends:

- Mach number increased downstream.
- Velocity increased downstream.
- Temperature decreased downstream.
- Pressure decreased downstream.

- Density decreased downstream.

These trends matched expected compressible flow theory.

Mass Flow Rate Estimation

Using the choked-flow mass flow equation:

$$\dot{m} = A^* P_0 \sqrt{\frac{\gamma}{RT_0}} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma + 1}{2(\gamma - 1)}}$$

Result:

- Mass Flow Rate = 39.48 kg/s

Thrust Estimation

Total thrust was estimated using:

$$F = \dot{m} V_e + (P_e - P_a) A_e$$

Results

- Momentum Thrust = 74.36 kN
- Pressure Thrust = -9.41 kN
- Total Thrust = 64.95 kN

Engineering Interpretation

The exit pressure was below atmospheric pressure:

- Exit Pressure = 44.7 kPa
- Atmospheric Pressure = 101.3 kPa

This produced a negative pressure-thrust contribution, indicating an overexpanded nozzle under the assumed operating conditions.

Despite this, the nozzle generated approximately 65 kN of total thrust.

Project Outcomes

Successfully completed:

- FreeCAD rocket nozzle design
- CAD geometry extraction

- Area-ratio analysis
- Numerical Area-Mach solver
- Mach reconstruction
- Temperature reconstruction
- Pressure reconstruction
- Density reconstruction
- Velocity reconstruction
- Mass flow estimation
- Thrust estimation

- Real combustion-product modeling
- CFD validation
- Particle-based flow visualization
- Thrust optimization studies

Conclusion

A complete 1D compressible-flow analysis framework was developed to evaluate the performance of a CAD-designed rocket nozzle. Using isentropic flow theory and numerical root solving, the project successfully reconstructed the nozzle flow field and estimated performance metrics including Mach number, velocity, mass flow rate, and thrust.

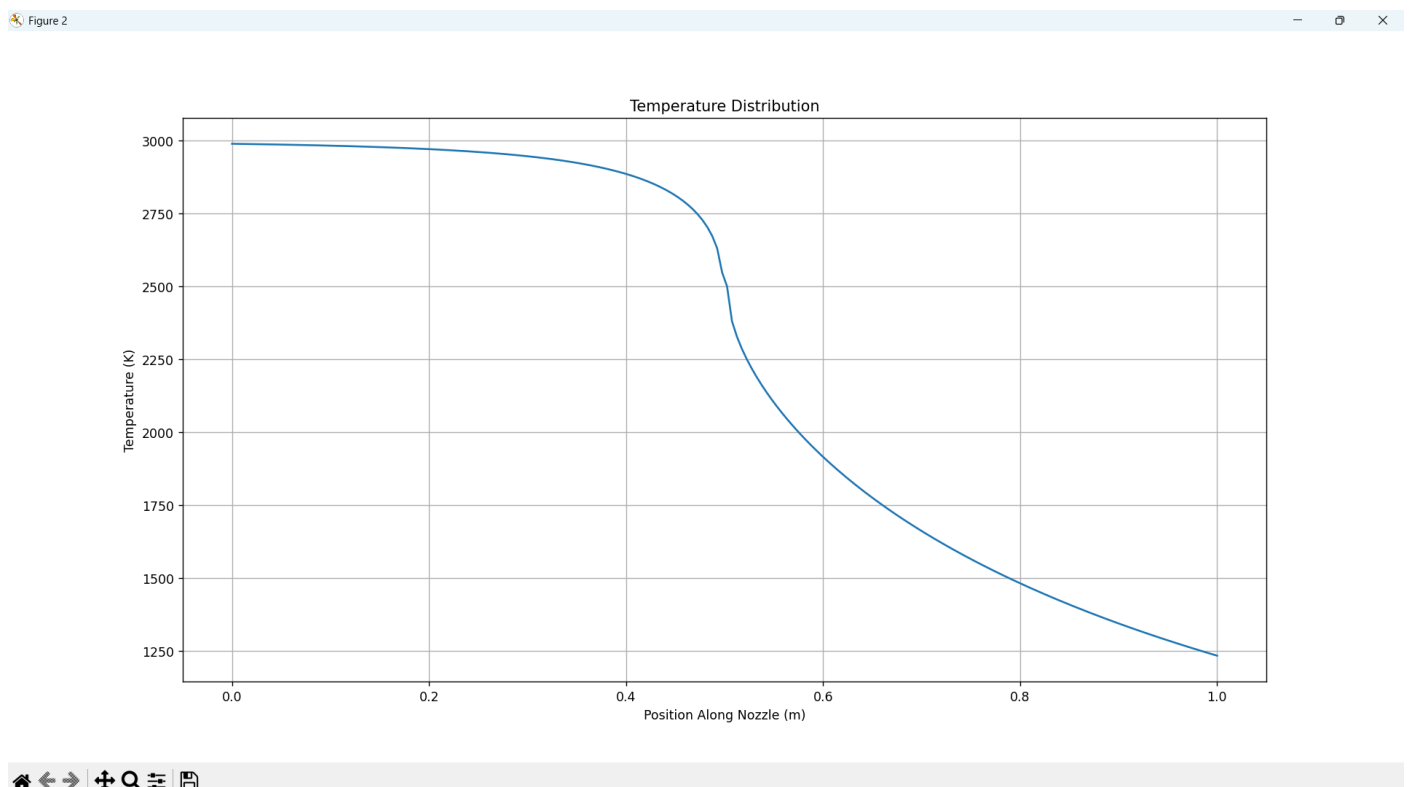
This work demonstrates the integration of CAD design, aerospace propulsion theory, numerical methods, and Python-based engineering analysis.

Future Work

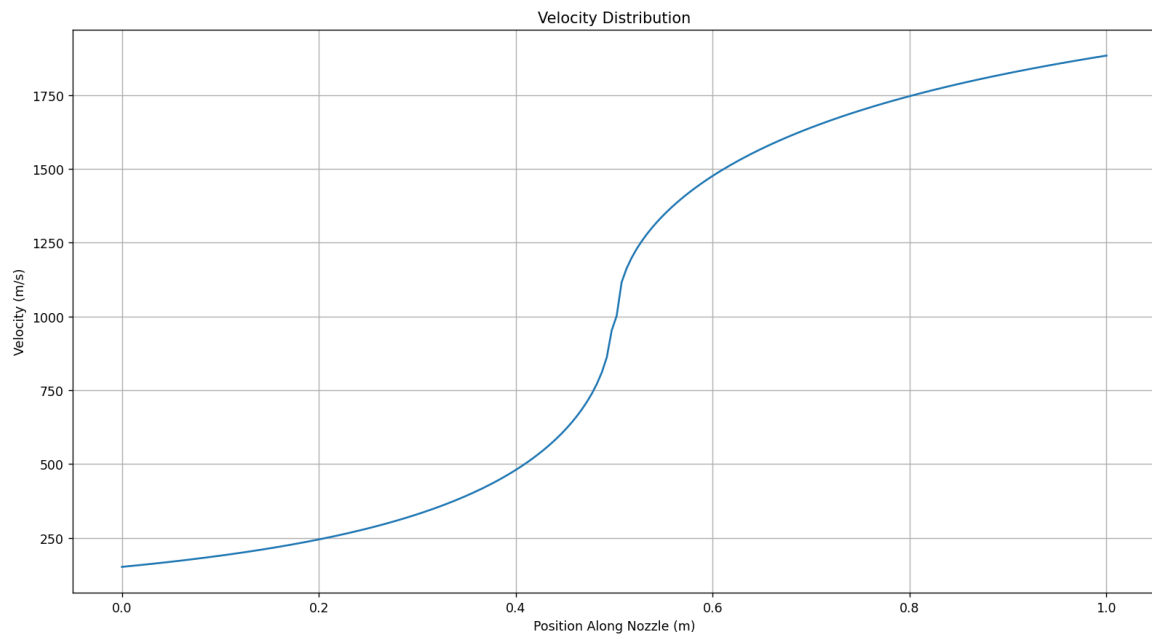
Potential extensions include:

- Importing full nozzle contour directly from FreeCAD
- Shock-wave modeling
- Variable gas properties

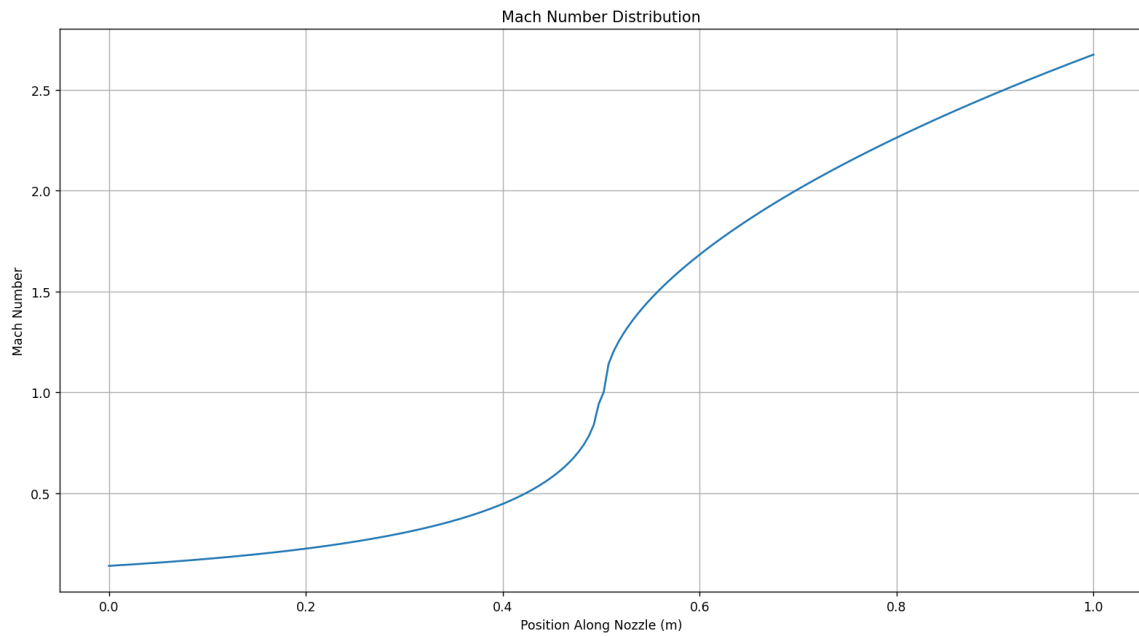
GRAPHS



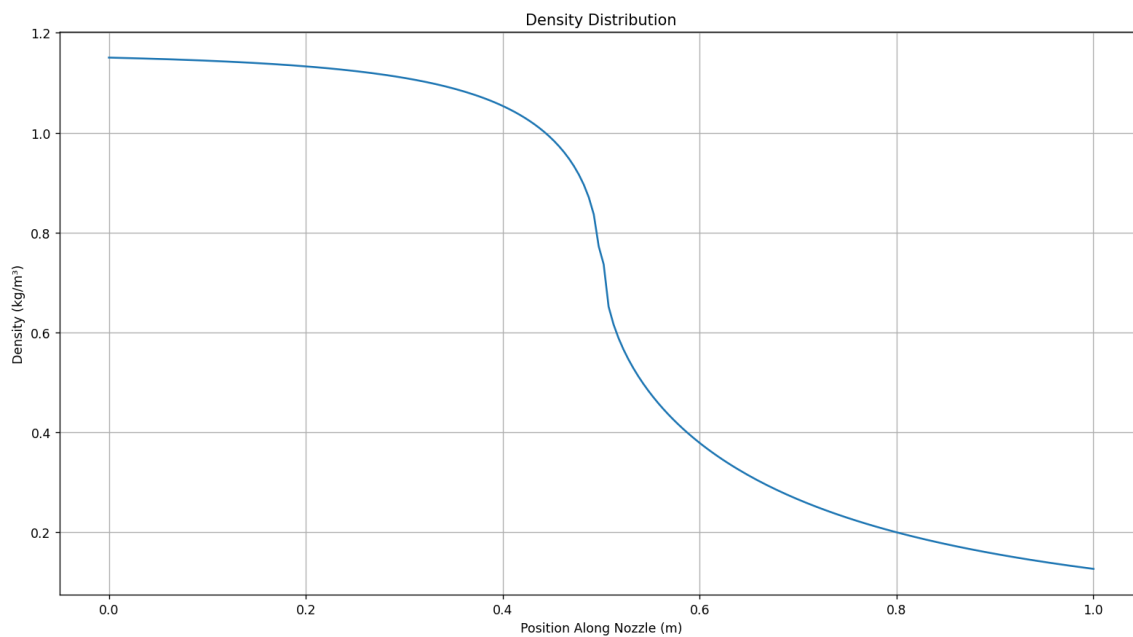
TEMPERATURE VARIATION GRAPH



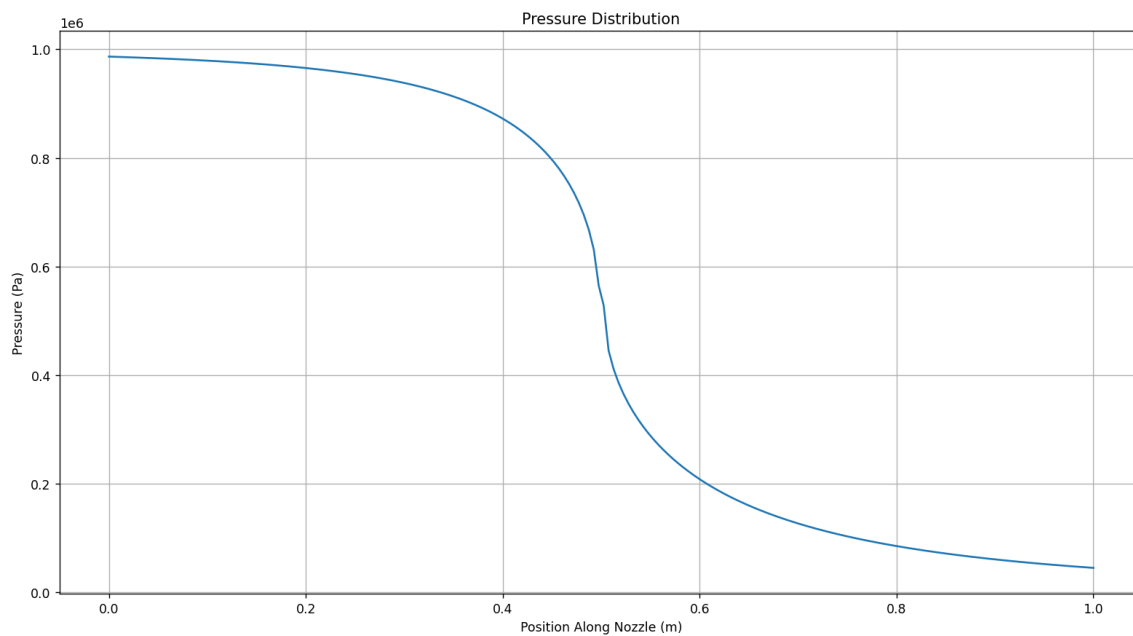
VELOCITY VARIATION GRAPH



MACH NUMBER VARIATION GRAPH



DENSITY VARIATION GRAPH



PRESSURE VARIATION GRAPH